

# Sara Klaasen

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## Education

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>ETH Zurich</b> , Geophysics <ul style="list-style-type: none"> <li>• PhD</li> <li>• <b>Thesis</b> distributed acoustic sensing for volcano monitoring</li> <li>• ETH medal</li> </ul> | Zurich, Switzerland<br>2020 – 2025     |
| <b>ETH Zurich</b> , Geophysics <ul style="list-style-type: none"> <li>• Msc</li> </ul>                                                                                                   | Zurich, Switzerland<br>2018 – 2020     |
| <b>UCR</b> , Liberal Arts and Sciences <ul style="list-style-type: none"> <li>• Bsc</li> <li>• Magna cum Laude</li> <li>• Earth Sciences, biology, mathematics and physics</li> </ul>    | Middelburg, Netherlands<br>2014 – 2017 |

## Experience

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Research School of Earth Sciences, Australian National University</b> , Postdoctoral researcher<br>Researching DAS projects in New Zealand. <ul style="list-style-type: none"> <li>• Imaging Taupo Volcano.</li> </ul> | Canberra, Australia<br>2026 – present<br>1 year |
| <b>ETH Zurich</b> , Postdoctoral researcher                                                                                                                                                                               | Zurich, Switzerland<br>2025 – 2026<br>1 year    |

## Awards

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| <b>ETH Medal</b><br>The ETH Medal is given to outstanding doctoral theses.<br>ETH Zurich<br><a href="https://ethz.ch/en/the-eth-zurich/education/awards/eth-medal.html">ethz.ch/en/the-eth-zurich/education/awards/eth-medal.html</a> | 2026 |
| <b>OSPA</b><br>Outstanding student presentation award<br>Seismological Society of America Photonics Meeting                                                                                                                           | 2024 |

## Grants

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| <b>SNF postdoc.mobility grant</b><br>A grant to fund a 2-year postdoc position.<br>Swiss National Science Foundation<br><a href="https://www.snf.ch/en/XIZpfY3iVS5KRRoD/funding/careers/postdoc-mobility">www.snf.ch/en/XIZpfY3iVS5KRRoD/funding/careers/postdoc-mobility</a> | 2025 |
| <b>VSbfonds grant</b><br>A grant to support a Masters programme abroad.<br>VSbfonds<br><a href="https://www.vsbfonds.nl">www.vsbfonds.nl</a>                                                                                                                                  | 2018 |

## Books

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### **Fiber optic sensing for volcano monitoring and imaging volcanic processes**

Jan 2025

Book chapter in Modern Volcano Monitoring

Jousset, Philippe and Currenti, Gilda and Murphy, Shane and Eibl, Eva and Caudron, Corentin and Retailleau, Lise and Wollin, Christopher and Klaasen, Sara and Nishimura, Takeshi and Fichtner, Andreas and Spica, Zack and Lemarchand, Arnaud and Krawczyk, Charlotte

[doi.org/10.1007/978-3-031-86841-2\\_17](https://doi.org/10.1007/978-3-031-86841-2_17)

### **The leading edge of volcanology: Advances and innovations in monitoring and interpreting volcanic activity**

Book chapter submitted to "The encyclopedia of Volcanoes, Third Edition"

Poland, Michael P and de Zeeuw-van Dalfsen, Elske and Aizawa, Koki and Klaasen, Sara and D'Arcy, Fiona and Nicholson, Emma J and Anderson, Kye

## Publications

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### **Submarine seismicity monitoring with distributed acoustic sensing near Santorini and Kolumbo Volcano**

2026

Sara Klaasen, Thomas Hudson, Sanchit Sachdeva, Paraskevi Nomikou, Andreas Fichtner

[doi.org/10.26443/seismica.v5i1.1845](https://doi.org/10.26443/seismica.v5i1.1845)

### **Subglacial volcano monitoring with fibre-optic sensing: Grímsvötn, Iceland**

2023

Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristín Jónsdóttir, Lars Gebraad, Patrick Paitz, Andreas Fichtner

[doi.org/10.30909/vol.06.02.301311](https://doi.org/10.30909/vol.06.02.301311)

### **Distributed acoustic sensing in volcano-glacial environments — Mount Meager, British Columbia**

2021

Sara Klaasen, Patrick Paitz, Nadja Lindner, Jan Dettmer, Andreas Fichtner

[doi.org/10.1029/2021JB022358](https://doi.org/10.1029/2021JB022358)

### **Seismic constraints on glacier density**

2025

Ariane Lanteri, Scott Keating, Lars Gebraad, Sara Klaasen, Marta Pienkowska-Cote, Olaf Eisen, Andrea Zunino, Kristín Jónsdóttir, Coen Hofstede, Dimitri Zigone, Andreas Fichtner

[doi.org/10.1038/s41598-025-26440-z](https://doi.org/10.1038/s41598-025-26440-z)

### **The Seismogenic Thickness of Venus**

2025

Julia Maia, Ana-Catalina Plesa, Iris van Zelst, Richard Ghail, Anna Gülcher, Mark Panning, Sven Peter Näsholm, Barbara de Toffoli, Anna Horleston, Krystyna Smolinski, Sara Klaasen, Robert Herrick, Raphael Garcia

[doi.org/10.1038/s41598-025-26440-z](https://doi.org/10.1038/s41598-025-26440-z)

### **Towards a widely applicable earthquake detection algorithm for fibreoptic and hybrid fibreoptic-seismometer networks**

2025

Thomas Hudson, Sara Klaasen, Olivier Fontaine, Conor Bacon, Kristin Jonsdottir, Andreas Fichtner

[doi.org/10.1093/gji/ggae459](https://doi.org/10.1093/gji/ggae459)

### **An illustrated guide to: Distributed and integrated fibre-optic sensing in seismology**

2024

Andreas Fichtner, Fabian Walter, Patrick Paitz, Sara Klaasen, Daniel Bowden, Sebastian Noe, Nils Müller, Dominik Husmann, Jacques Morel

[dx.doi.org/10.1016/j.eqs.2024.09.006](https://dx.doi.org/10.1016/j.eqs.2024.09.006)

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| <b>Seismic wave detectability on Venus using ground deformation sensors, infrasound sensors on balloons and airglow imagers</b><br>Raphael Garcia, Iris van Zelst, Taichi Kawamura, Sven Peter Näsholm, Anna Horleston, Sara Klaasen, Maxence Lefevre, Celine Marie Solberg, Krystyna Smolinski, Ana-Catalina PLesa, Quentin Brissaud, Julia Maia, Simon Stähler, Philippe Lognonné, Mark Panning, Anna Gülcher, Richard Ghail, Barbara de Toffoli<br><a href="https://doi.org/10.1029/2024EA003670">doi.org/10.1029/2024EA003670</a> | 2024 |
| <b>Modelling uncertainty in P-wave arrival-times retrieved from DAS data: case-studies from 15 fibre optic cables</b><br>Emanuele Bozzi, Nicola Piana Agostinetti, Andreas Fichtner, Sara Klaasen, Arantza Ugalde, Biondo Biondi, S Yuan, Torsten Dahm, Marius Isken, Patrick Paitz, Fabian Walter, Alain Baird, Carlos Becerril, Takeshi Nishimura, Junzhu Shen, Tieyuan Zhu, Gilberto Saccorotti<br><a href="https://doi.org/10.1093/gji/ggae364">doi.org/10.1093/gji/ggae364</a>                                                   | 2024 |
| <b>Complex spatial distribution of onset amplitude and waveform correlation: case studies from different DAS experiments</b><br>Emanuele Bozzi, Gilberto Saccorotti, Nicola Piana Agostinetti, Carlos Becerril, Andreas Fichtner, Sara Klaasen, Takeshi Nishimura, Junzhu Shen, Fabian Walter, Tieyuan Zhu<br><a href="https://doi.org/10.4430/bgo00458">doi.org/10.4430/bgo00458</a>                                                                                                                                                 | 2024 |
| <b>Challenges in submarine fiber-optic earthquake monitoring</b><br>Jonas Igel, Sara Klaasen, Sebastian Noe, Paraskevi Nomikou, Konstantinos Korantzalos, Andreas Fichtner<br><a href="https://doi.org/10.1029/2024JB029556">doi.org/10.1029/2024JB029556</a>                                                                                                                                                                                                                                                                         | 2024 |
| <b>Fiber-optic observation of volcanic tremor through floating ice sheet resonance</b><br>Andreas Fichtner, Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Patrick Paitz, Kristin Jonsdottir<br><a href="https://doi.org/10.1785/0320220010">doi.org/10.1785/0320220010</a>                                                                                                                                                                                                                                                    | 2022 |
| <b>Detecting Seismic Events with Computer Vision: Applications for Fiber-Optic Sensing</b><br>Solvi Thrastarson, Robert Torfason, Sara Klaasen, Patrick Paitz, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner<br><a href="https://doi.org/10.1002/essoar.10509693.1">doi.org/10.1002/essoar.10509693.1</a>                                                                                                                                                                                                                 | 2021 |

## Skills

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**Physics**

## Languages

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**Dutch**

native

**English**

C2

**German**

B2

**French**

B1

## Interests

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**Geophysics**